

---

# Information flows, but can you catch it?

---

**Vlad Vladutu**

Aarhus University  
au799236@uni.au.dk

**Arne Wiese**

Aarhus University  
202500860@post.au.dk

## Abstract

Podcasts influence public narratives, but their volume makes systematic analysis difficult. Using the Structured Podcast Research Corpus (SPoRC) and semantic sentence embeddings, we map episode transcripts into a shared representation space and derive time series of theme (“idea”) intensity from embedding similarities. We construct similarity graphs at the episode, podcast, and host levels to study community structure and cross-category connectivity. To explore links between textual themes and external signals, we compare idea intensity with stock market indicators using Pearson and Spearman correlations (reporting p-values) and complementary Granger-causality tests on aligned time series. Given the limited time horizon and sensitivity to similarity thresholding and lag selection, evidence for robust lead-lag relationships is weak, but the embedding-based pipeline yields interpretable clusters and cross-category bridges that help navigate large podcast collections and generate testable hypotheses about how themes move across shows and domains.

## 1 Introduction

Podcasts are now a major channel for long-form information, with broad reach and low barriers to publication [Wirtschafter, 2023]. Their influence is amplified by the fact that claims and narratives can spread widely without passing through traditional editorial filters, yet the ecosystem remains difficult to study at scale because listening does not scale.

New corpora and NLP tools make ecosystem-level analysis feasible. The Structured Podcast Research Corpus (SPoRC) provides transcripts and metadata for a large snapshot of English-language podcasts (May–June 2020), enabling comparative analysis across shows, categories, and hosts [Litterer et al., 2024]. Building on this, we ask a content-focused question: *how do ideas propagate through podcasts?* Which shows pick up which ideas, how strongly do they appear, and how are these patterns organized?

We embed transcripts into a shared semantic space to extract and track ideas, then construct similarity graphs to map connections between episodes, podcasts, and hosts. Finally, we explore whether idea intensity aligns with financial market dynamics via exploratory causality tests. Our goal is a compact, navigable “map” of podcast content that supports studying thematic flow across time and domains.

## 2 Related Work

### 2.1 SPoRC Dataset

Litterer et al. [2024] introduce SPoRC, a “thick sample” of English-language podcasts from May–June 2020 built from public RSS feeds. They collect audio and metadata for 1.3M episodes from 247K shows; after filtering for transcription quality, they release 1.1M transcripts (6.6B words). SPoRC includes episode metadata and inferred host/guest identities; for 370K episodes it also provides speaker turns and audio-derived features. We use SPoRC in all experiments.

## 2.2 Podcast networks and topical responsiveness

Using SPoRC, Litterer et al. [2024] analyze topical structure and a guest co-appearance network, and study temporal responsiveness via a George Floyd case study (attention increases then decays). Their results motivate our focus on linking content across episodes, but their analyses do not directly support user-defined concepts or passage-level semantic matching.

## 2.3 Sentence-BERT

Reimers and Gurevych [2019] show that BERT cross-encoders are accurate for sentence-pair scoring but costly for retrieval because each query–candidate pair must be encoded jointly. SBERT fine-tunes a Transformer in a siamese (optionally triplet) setup to produce fixed-size embeddings compared via cosine similarity, enabling efficient semantic search and clustering.

## 2.4 T5

Raffel et al. [2023] propose T5, which casts NLP tasks into a text-to-text format: an encoder–decoder Transformer generates targets from an input string (often with a task prefix). T5 is pre-trained on C4 using span-corruption denoising and then fine-tuned on downstream text-to-text tasks. We use T5 to generate short textual descriptions from top retrieved passages.

# 3 Approach

## 3.1 Data Preprocessing

SPoRC provides extensive metadata and auxiliary annotations (Section 2.1), but many fields are not directly useful for content-based retrieval. Several speaker-level voice metrics are reported only as turn-level averages, which are weak indicators of within-episode variation. To reduce noise and keep the pipeline reproducible, we restrict inputs to fields used to (i) identify episodes, (ii) group them over time, and (iii) apply coarse topic filters: {transcript, podTitle, epTitle, episodeDateLocalized, hostPredictedNames, guestPredictedNames, category1, category2, category3}.

We retain up to three category labels to enable deterministic filtering (via a small whitelist/blacklist), removing broad categories that can introduce shared background signal and increase spurious similarity between unrelated episodes.

We also inspect corpus coverage over time (Figure 1). Days with unusually low episode counts are excluded from market-alignment analyses to avoid artifacts from missing data. For remaining days, we sample a fixed number of episodes per day using a fixed random seed to equalize daily weight in time-series aggregation; varying the seed serves as a robustness check.

## 3.2 Text Embedding Strategies

We map each episode to a vector representation and compare episodes (or queries vs. episodes) using cosine similarity. Because transcripts may exceed model context limits, we chunk each transcript into fixed-length token windows (256 tokens per chunk) with overlap (fixed stride of 32) to reduce boundary effects. Chunk embeddings are computed in GPU batches and cached to disk, enabling evaluation of multiple aggregation strategies without re-encoding.

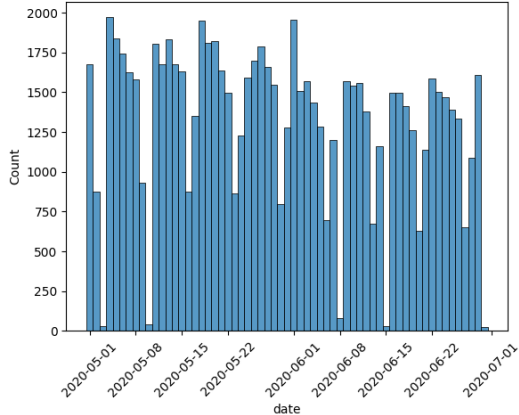
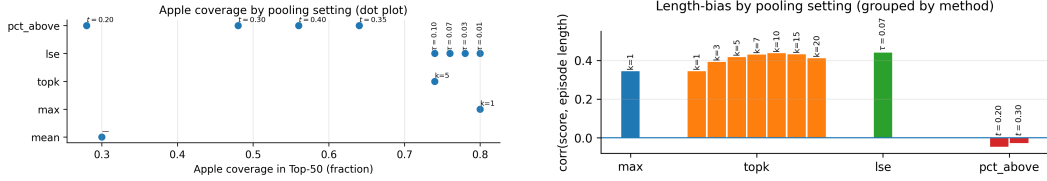


Figure 1: Daily episode counts in SPoRC before preprocessing. Days with unusually low coverage are excluded from market-alignment experiments to avoid artifacts from missing or sparse data.



(a) Entity-query validation on an Apple description: coverage in the top-50 retrieved episodes and mean mention count, comparing aggregation methods/hyperparameters.

(b) Correlation between idea scores and episode length (tokens) across aggregation methods. High correlation indicates length bias; whether this is desirable depends on whether episodes should contribute equally or proportionally to available content.

Figure 2: Idea-scoring validation and bias checks across aggregation methods.

### 3.2.1 Models and pooling.

Our final experiments use `sentence-transformers/all-MiniLM-L6-v2`. During preliminary development we also tested `BAAI/bge-small-en-v1.5`, but we discontinued it and exclude those runs from the results presented here. We used the standard attention-mask mean pooling for the MiniLM Sentence-Transformers.

### 3.2.2 Episode-level aggregation.

Given chunk embeddings (or query-chunk similarity scores), we aggregate across chunks to obtain a single episode score. We evaluate mean and max aggregation, and additionally some idea-aware methods: (i) log-mean-exp (smooth max; with varying temperature), (ii) top- $k$  mean (varying  $k$ ), and (iii) percent-above-threshold (varying threshold), which measures how broadly an idea is supported across the episode rather than by a single spike.

## 4 Experiments

### 4.1 Idea extraction

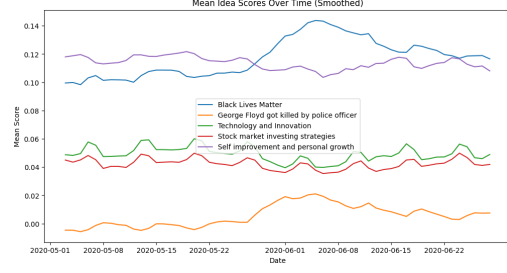
A core operation in our pipeline is estimating how strongly an episode expresses a target idea. For each idea, we embed a short description (e.g., “*Self Improvement and Personal Development*”) and compute cosine similarity against every transcript-chunk embedding. We score at the chunk level (rather than using a single pre-pooled episode embedding) so that aggregation can control whether an idea is supported broadly across the episode or only by a few salient segments.

All experiments use the filtered episode set (category whitelisting and standardized per-day sampling; Section 3.1). We validate scoring in three ways. First, we inspect the top-ranked episode per idea; Figure 3a visualizes the highest-scoring transcript as a qualitative sanity check. Second, we quantify entity-query retrieval precision: given a firm description (e.g., “*Apple Inc. AAPL (not fruit): Cupertino tech firm; iPhone/iOS, Mac ...*”), we report (i) the fraction of the top-50 retrieved episodes that mention the firm (simple string matching) and (ii) the mean number of mentions, comparing aggregation methods and hyperparameters (Figure 2a). As a lexical sanity-check baseline, we rank episodes by TF-IDF cosine similarity between the firm description and episode text and evaluate with the same mention-based metrics. Across entity queries, `topk_mean` with  $k=1$  achieves similar precision@50 and mean-mentions@50 to TF-IDF; we treat these comparisons as descriptive (no significance testing). We find  $k=1$  performs best for entity queries, consistent with firm discussion being concentrated in a small number of highly relevant segments, while larger  $k$  can dilute the signal. Third, we assess separability and bias: we compute cosine similarity between *idea-description embeddings* to confirm that selected ideas are distinct in embedding space (Figure 4), and we correlate per-episode idea scores with episode length (tokens) to diagnose length-driven inflation (Figure 2b). (Correlations between daily idea-score time series are reported in the Appendix.)

Beyond ranking, we aggregate episode scores into daily time series by averaging over episodes per day (Figure 3b). This yields an *idea intensity* signal used in downstream temporal analyses.



(a) Word cloud of the highest-scoring episode.



(b) Daily mean idea score over time.

Figure 3: Idea extraction results for *Self Improvement and Personal Development*.

## 4.2 Idea generation

Idea extraction starts from a user-specified idea; idea generation produces a short, headline-style statement from an episode. We use it sparingly as a qualitative aid: (i) to summarize retrieved evidence, and (ii) to optionally propose candidate ideas that can be fed back into the extraction pipeline. We implement evidence-grounded generation with `flan-t5-large`: we first select a small set of representative sentences, then generate one short statement constrained to use only that evidence.

### 4.2.1 Evidence selection (MMR).

We select  $k$  sentences with Maximal Marginal Relevance (MMR), balancing centrality and non-redundancy:

$$\lambda \cdot \text{sim}(s, c) - (1 - \lambda) \cdot \max_{s' \in S} \text{sim}(s, s'),$$

where  $c$  is the centroid of sentence embeddings for the episode,  $S$  is the set of already selected sentences,  $\text{sim}$  is cosine similarity, and  $\lambda = 0.7$ .

### 4.2.2 Prompt.

We use a compact, constraint-based prompt that enforces a single short sentence and returns `unclear` when the evidence is insufficient or off-topic. For the episode shown in the word cloud (Figure 3a), the model outputs: “*The brain is a complex system that is constantly changing and adapting.*” This sentence serves as an extracted core idea and is more specific than the high-level terms in the word cloud. The extracted idea can then be fed back into the pipeline to retrieve more concrete matches than those obtained through simple semantic similarity alone.

## 4.3 Graph construction and visualization

To make the embedding space interpretable, we build sparse similarity graphs where nodes are items (episodes, podcasts, or hosts) and edges encode semantic proximity. For each item type, we add an undirected edge  $(i, j)$  only if  $\text{sim}(e_i, e_j) \geq \tau = 0.2$ . We choose  $\tau$  empirically to balance connectivity and readability: lower values produce a more crowded graph, while higher values sparsify it. Within a reasonable neighborhood of the chosen  $\tau$ , the graph’s qualitative patterns remain stable. To support shortest-path queries, we use an edge *length* of  $w_{ij} = 1/\text{sim}(e_i, e_j)$ , so stronger semantic links correspond to shorter paths.

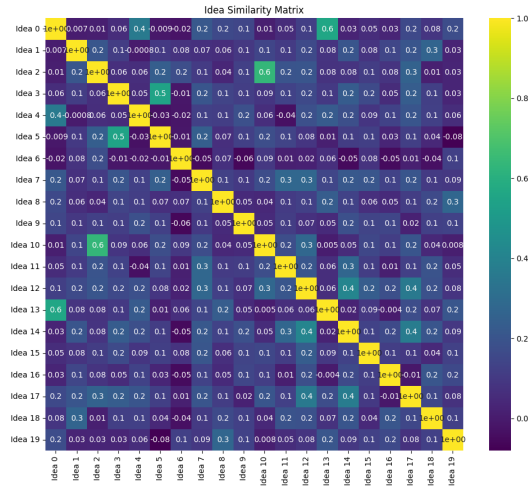


Figure 4: Cosine similarity between idea-description embeddings for 20 precise ideas (May–June). Lower off-diagonal values indicate more distinct idea signals.

Figure 5 shows the episode graph for a random sample of 5,000 episodes; nodes are colored by `category1` as a proof that embedding neighborhoods align with coarse labels. We construct analogous podcast- and host-level graphs and complementary tables (Appendix A).

#### 4.4 Correlating ideas with stock returns

To test whether podcast discourse precedes market movements, we evaluate whether daily *idea intensity* helps predict asset returns. Stock data are fetched from Yahoo Finance Aroussi [2025]. For each instrument, we compute daily log returns and volume and align them with daily aggregated idea scores. We then run Granger causality tests (via `statsmodels` Perktold et al. [2024]) and compute Pearson and Spearman correlations with p-values. For lag  $k$ , the null hypothesis is that lagged idea scores do not improve prediction of today’s return beyond past returns; we compare a returns-only autoregressive model against one that additionally includes lagged idea scores, using an F-test for incremental explanatory power. These results are exploratory associations and do not imply causal market impact.

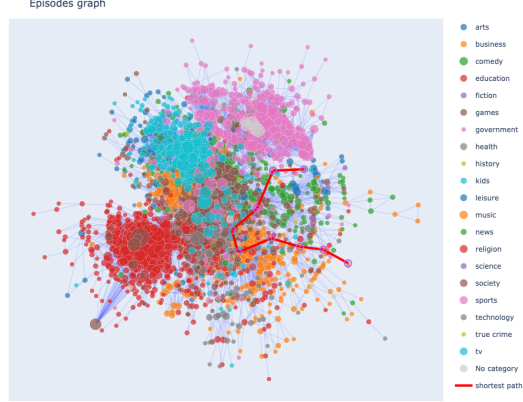


Figure 5: Episode similarity graph on a sample of 5,000 episodes. Nodes are colored by `category1`; edges connect pairs with cosine similarity  $\geq \tau$  (weight =  $1/\cos$ ).

## 5 Analysis

### 5.1 Fine-grained events vs. general ideas

The idea-intensity series often exhibits sharp spikes followed by slower decay, especially for event-linked ideas, resembling the *media-storm* pattern reported for news coverage [Boydston et al., 2014]. In contrast, broader ideas are easier to match across episodes and therefore tend to yield higher similarity scores, but they are less specific and are better suited for coarse filtering than fine-grained analysis. Although podcasts differ from news in format and cadence, this pattern is consistent with collective responsiveness to major events. Since the signal is derived from aggregated embedding similarities rather than explicit event labels, we treat these dynamics as suggestive rather than conclusive.

### 5.2 Idea extraction and generation

The extraction results indicate that embedding-based similarity captures topical relevance beyond keyword overlap: for “*Self Improvement and Personal Development*”, the top-ranked episode contains terms aligned with the intended theme, consistent with prior findings on sentence embeddings [Reimers and Gurevych, 2019]. The generation experiment (Flan-T5) produces coherent, evidence-grounded statements, but the outputs are often generic, suggesting a bias toward safe generality over episode-specific detail. This limits usefulness for fine-grained analysis without stronger constraints and/or more selective evidence extraction.

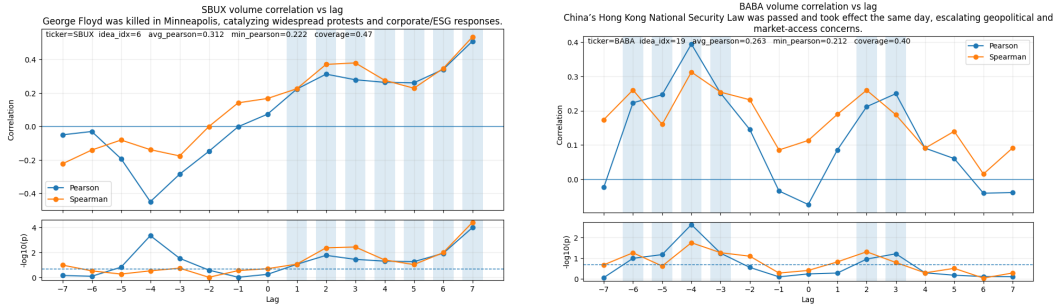
### 5.3 Communities based on graphs and ideas

The similarity graphs reveal both coherent category-based clusters and a non-trivial number of cross-category connections. The clustering suggests that semantic embeddings recover much of the dataset’s coarse organization, consistent with prior analyses of SPoRC [Litterer et al., 2024]. Cross-category edges highlight that many ideas cut across labels, which is desirable for studying information flow but complicates interpretation. Modularity and density statistics support these observations, but they depend on sampling and the similarity threshold; we therefore treat them as descriptive summaries rather than stable properties of the full ecosystem.

## 5.4 Stock market association

Across assets and ideas, Granger tests and lagged correlations occasionally yield statistically significant results at specific lags, but the dominant pattern is that idea intensity more often follows market movements than precedes them. Given the short time window, multiple hypothesis testing, and exploratory setup, we avoid strong predictive claims. Moreover, Granger causality does not imply true causal influence and may reflect shared responses to external events. We manually inspected many of our results, as there are no clear baselines.

When scanning many idea–ticker pairs (including fine-grained event ideas, broader ideas, and T5-generated candidates), we observed that trading *volume* exhibits more frequent and stable associations than returns. Aggressive aggregation (e.g., top- $k$  mean with  $k = 1$ ) can amplify isolated high-similarity chunks and produces spurious lead–lag patterns (Figure 6, left). We therefore apply a conservative screening heuristic: retain only cases with Pearson p-values  $|p| \leq 0.2$  and qualitatively similar Spearman correlation, and require the effect to persist across at least five lags (ideally consecutive). Figure 6 illustrates both a spurious example under aggressive pooling (SBUX) and a case where podcast discussion increases *after* a market move (BABA), consistent with a reactive rather than predictive signal.



(a) Aggressive pooling (top- $k$  mean,  $k = 1$ ) yields a misleading correlation pattern (SBUX vs. an event-linked idea).

(b) Percent-above-threshold pooling ( $\tau = 0.3$ ) shows increased discussion *after* a BABA move (China–US tension idea), consistent with a reactive effect.

Figure 6: Lag-scan correlations over lags  $-7$  to  $+7$  days. A  $\pm 7$ -day window is reasonable because podcasts often discuss topics on a weekly/every 2 weeks cadence, and two weeks is a meaningful but not overbroad slice of the full 2-month analysis period. Highlighted regions indicate lags passing the screening heuristic.

## 6 Conclusion

We investigated how large-scale podcast transcripts from SPoRC can be analyzed with modern NLP methods to characterize themes and semantic structure. Using transformer embeddings with chunk-level scoring, we extracted and (qualitatively) generated headline-style ideas, and built thresholded similarity graphs over episodes and metadata entities. Across these views, semantic representations largely recover the dataset’s coarse topical organization while also surfacing cross-category links that are not captured by labels alone. We also aligned daily aggregated idea intensity with market variables (returns and volume) to run exploratory lead–lag analyses; in most cases, the dominant pattern was reactive rather than predictive.

Overall, the pipeline illustrates how embedding-based representations can support navigation and lightweight analysis of large, unstructured podcast corpora. Key limitations include the short time window, sensitivity to pooling/threshold choices, and limited ground-truth validation, motivating more rigorous evaluation and robustness testing in future work.

## References

- Ran Aroussi. yfinance: Download market data from yahoo! finance, 2025. URL <https://pypi.org/project/yfinance/>. Python package (PyPI). Accessed: 2025-12-16.
- Amber E. Boydston, Anne Hardy, and Stefaan Walgrave. Two faces of media attention: Media storm versus non-storm coverage. *Political Communication*, 31(4):509–531, 2014. doi: 10.1080/10584609.2013.875967. URL <https://doi.org/10.1080/10584609.2013.875967>.
- Benjamin Litterer, David Jurgens, and Dallas Card. Mapping the podcast ecosystem with the structured podcast research corpus, 2024. URL <https://arxiv.org/abs/2411.07892>.
- Josef Perktold, Skipper Seabold, Kevin Sheppard, Chad Fulton, Kerby Shedden, J. Brockmendel, J. Grana, Peter Quackenbush, Vincent Arel-Bundock, Wes McKinney, Ian Langmore, Bart Baker, Ralf Gommers, Yogabonito, S. Scherrer, Yauhen Zhurko, Matthew Brett, Enrico Giampieri, Yl565, and Yaroslav Halchenko. statsmodels/statsmodels: Release 0.14.2, 2024. URL <https://doi.org/10.5281/zenodo.10984387>.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer, 2023. URL <https://arxiv.org/abs/1910.10683>.
- Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks, 2019. URL <https://arxiv.org/abs/1908.10084>.
- Valerie Wirtschafter. Audible reckoning: How top political podcasters spread unsubstantiated and false claims. Brookings, March 2023. URL <https://www.brookings.edu/articles/audible-reckoning-how-top-political-podcasters-spread-unsubstantiated-and-false-claims/>. Accessed: 2024-12-02.

## A Appendix for graphs

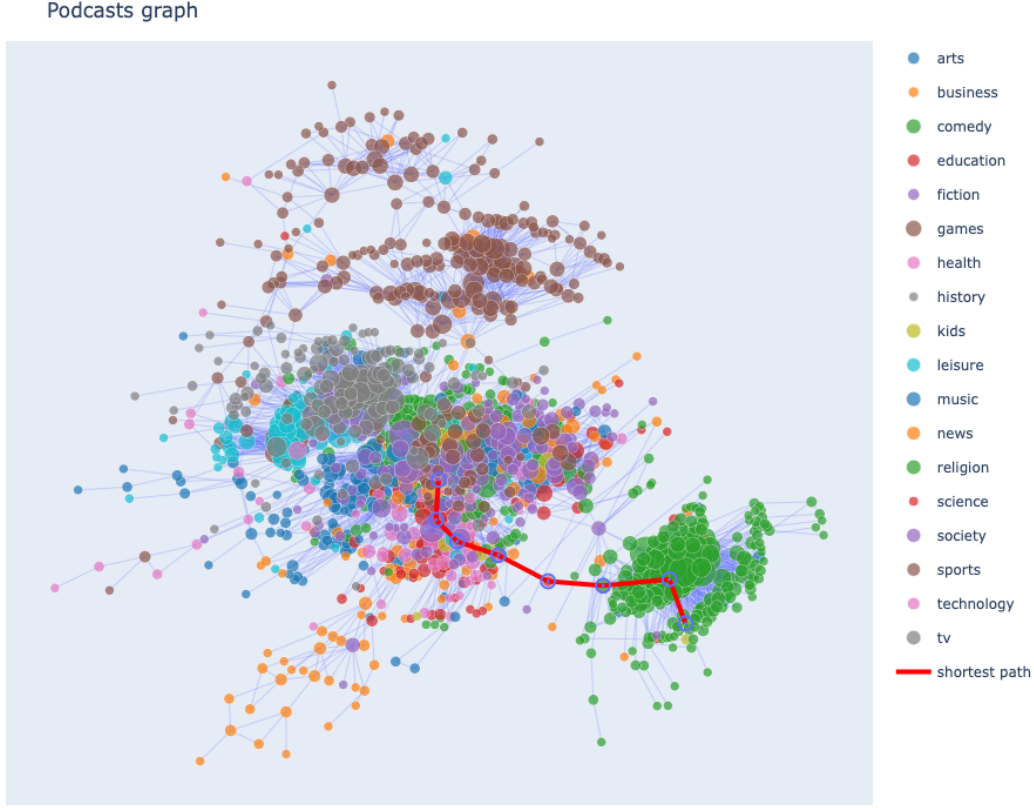


Figure 7: Podcast-level similarity graph built from aggregated transcript embeddings using mean. Nodes represent podcasts (colored by category1); edges connect podcast pairs with cosine similarity  $\geq \tau = 0.2$  under the same embedding/aggregation pipeline as the episode graph. The highlighted path illustrates an example shortest semantic connection between two selected podcasts.

Table 1: Category modularity contribution (5,000-sample graphs).

| Category  | Episodes     | Podcasts     | Hosts        |
|-----------|--------------|--------------|--------------|
| Arts      | –            | 0.014        | –            |
| Business  | 0.065        | 0.010        | 0.052        |
| Comedy    | 0.083        | 0.099        | 0.032        |
| Education | 0.022        | –            | 0.017        |
| Health    | 0.026        | 0.015        | –            |
| Leisure   | 0.015        | 0.063        | 0.034        |
| Music     | 0.036        | 0.047        | 0.039        |
| News      | –            | –            | 0.031        |
| Religion  | <b>0.323</b> | <b>0.381</b> | 0.100        |
| Society   | 0.042        | 0.042        | 0.029        |
| Sports    | 0.212        | 0.193        | <b>0.496</b> |
| TV        | 0.086        | 0.140        | 0.193        |

Table 2: Category internal density (5,000-sample graphs).

| Category  | Episodes     | Podcasts     | Hosts        |
|-----------|--------------|--------------|--------------|
| Arts      | –            | 0.018        | –            |
| Business  | 0.015        | 0.013        | 0.011        |
| Comedy    | 0.058        | 0.043        | 0.045        |
| Education | 0.016        | –            | 0.019        |
| Health    | 0.025        | 0.033        | –            |
| Leisure   | 0.034        | 0.059        | 0.063        |
| Music     | <b>0.158</b> | <b>0.096</b> | <b>0.115</b> |
| News      | –            | –            | 0.020        |
| Religion  | 0.043        | 0.058        | 0.029        |
| Society   | 0.018        | 0.022        | 0.013        |
| Sports    | 0.050        | 0.036        | 0.037        |
| TV        | 0.062        | 0.070        | 0.078        |

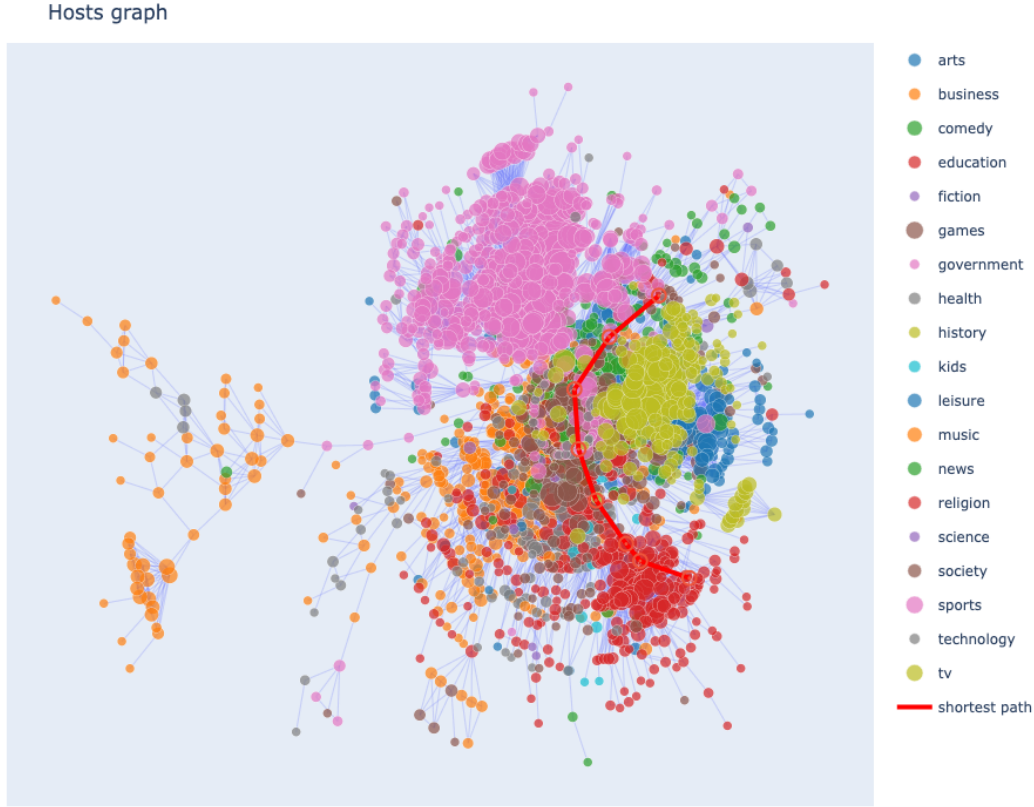


Figure 8: Host-level similarity graph constructed from aggregated host embeddings derived from their associated episodes(same as podcast, it uses mean as aggregation). Nodes represent hosts (colored by category), edges indicate host pairs with cosine similarity  $\geq \tau = 0.2$ . The highlighted path shows an example shortest semantic link between two selected hosts.

## B Appendix for idea generation

Idea: Write ONE factual, headline-style sentence using ONLY the evidence.  
 Constraints:  $\leq 24$  words; 1 sentence; no pronouns; no opinions. Do not mention podcast/show/speaker; paraphrase (no quotes/titles). If evidence is insufficient or mostly meta/filler: unclear. <evidence> - ... (evidence items inserted here) ... </evidence> output:

Listing 1: T5 prompt used for idea generation

## C Appendix for idea extraction

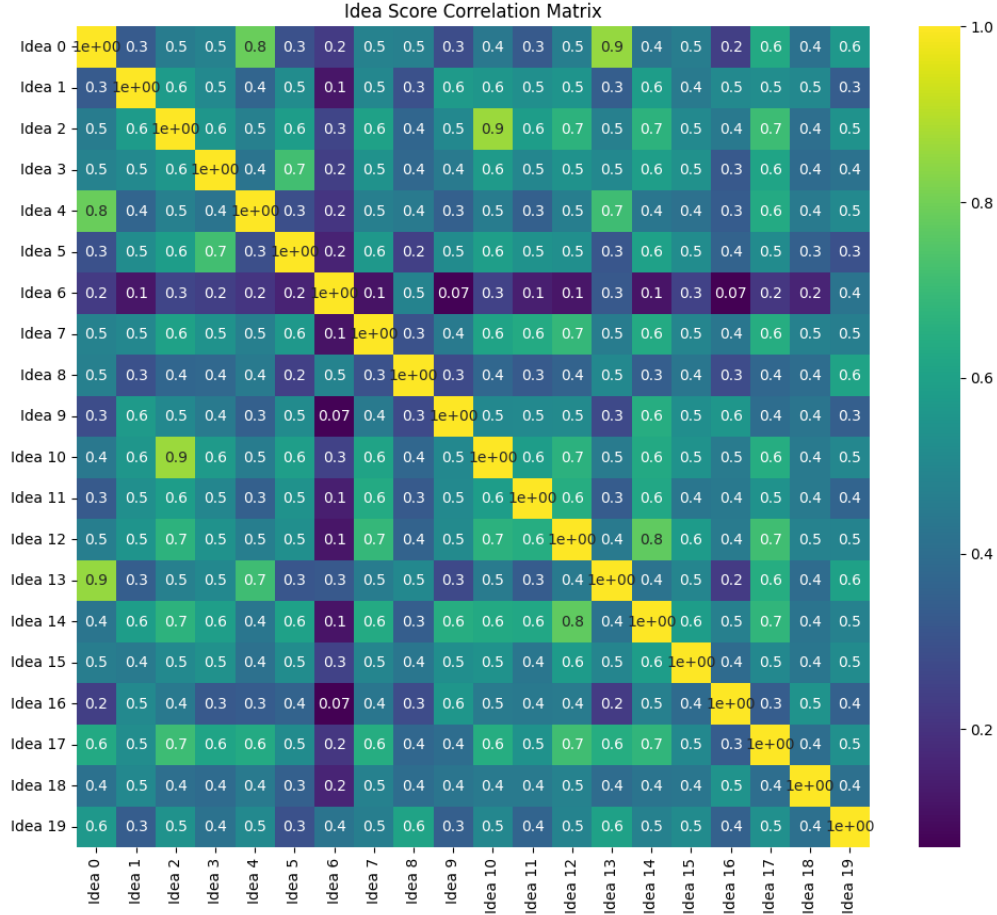


Figure 9: **Pairwise Pearson correlations between the daily idea-score time series** for the 20 fine-grained ideas (topk\_mean aggregation,  $k = 1$ ). The mean off-diagonal correlation is 0.457, indicating that while the idea descriptions are relatively distinct in embedding space (cf. Figure Y), their temporal dynamics across days are substantially more aligned.

| Company   | Company description query                                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apple     | Apple Inc. AAPL (not fruit): Cupertino tech firm; iPhone/iOS, Mac, iPad; Apple Watch; App Store; revenue, earnings, market cap                        |
| Google    | Google LLC: Mountain View tech giant; Search, Ads, Android, YouTube; Cloud services; AI research; revenue, earnings, market cap                       |
| Microsoft | Microsoft Corporation MSFT: Redmond software behemoth; Windows OS, Office Suite; Azure cloud; LinkedIn; gaming (Xbox); revenue, earnings, market cap  |
| Amazon    | Amazon.com Inc. AMZN: Seattle e-commerce titan; online retail, AWS cloud services; Prime membership; logistics network; revenue, earnings, market cap |

| ID | Idea (short description)                                                                                                                               | Indicative date | Possibly related tickers   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------|
| 0  | The FDA issued an Emergency Use Authorization (EUA) for remdesivir for severe COVID-19 cases, boosting antiviral focus.                                | 2020-05-01      | GILD, MRNA, XLV, IBB       |
| 1  | At Berkshire Hathaway’s annual meeting, Warren Buffett said Berkshire fully exited its major U.S. airline stakes, signaling prolonged aviation stress. | 2020-05-02      | AAL, DAL, UAL, LUV         |
| 2  | The U.S. April jobs report showed a historic collapse (20.5M jobs lost; unemployment 14.7%), driving macro volatility.                                 | 2020-05-08      | SPY, IWM, TLT, XLF         |
| 3  | JCPenney filed for Chapter 11 bankruptcy protection amid pandemic-driven retail pressure.                                                              | 2020-05-15      | SPG, MAC, XRT, M           |
| 4  | Moderna announced positive interim Phase 1 data for its mRNA-1273 COVID-19 vaccine candidate, sparking a biotech rally.                                | 2020-05-18      | MRNA, IBB, PFE, JNJ        |
| 5  | Hertz filed for Chapter 11 bankruptcy after a sharp demand collapse in travel and car rentals.                                                         | 2020-05-22      | HTZ, CAR, JETS, AAL        |
| 6  | George Floyd was killed in Minneapolis, catalyzing widespread protests and corporate/ESG responses.                                                    | 2020-05-25      | NKE, SBUX, WMT, TGT        |
| 7  | The European Commission presented its recovery plan proposal, shifting expectations for EU-wide fiscal support.                                        | 2020-05-27      | VGK, EZU, FXE, DB          |
| 8  | President Trump signed Executive Order 13925 on online censorship, raising perceived regulatory risk for social platforms.                             | 2020-05-28      | TWTR, META, GOOGL, SNAP    |
| 9  | Nikola began trading under ticker NKLA following its SPAC merger, intensifying EV/speculative trading activity.                                        | 2020-06-04      | NKLA, TSLA, QS, GM         |
| 10 | The U.S. May jobs report surprised with +2.5M payrolls and unemployment down to 13.3%, shifting reopening narratives.                                  | 2020-06-05      | SPY, IWM, XLI, XLY         |
| 11 | OPEC+ agreed to extend record oil production cuts, supporting crude prices and energy credit.                                                          | 2020-06-06      | XOM, CVX, XLE, USO, SLB    |
| 12 | The Fed released its June FOMC statement and projections, reinforcing near-zero rates and shaping forward guidance.                                    | 2020-06-10      | TLT, IEF, SPY, GLD         |
| 13 | The FDA revoked the EUA for hydroxychloroquine and chloroquine for COVID-19, after safety/efficacy reassessment.                                       | 2020-06-15      | SNY, PFE, GILD, XLV        |
| 14 | The New York Fed announced the SMCCF would begin buying corporate bonds (beyond ETFs), tightening credit spreads.                                      | 2020-06-15      | LQD, HYG, JNK, SPY         |
| 15 | Wirecard disclosed indications that €1.9B was missing and delayed its financial statements, triggering a major fraud scandal.                          | 2020-06-18      | V, MA, PYPL, SQ            |
| 16 | Apple announced the Mac transition to Apple silicon at WWDC, reshaping semiconductor and PC ecosystem expectations.                                    | 2020-06-22      | AAPL, TSM, INTC, AMD, NVDA |
| 17 | The Federal Reserve released 2020 stress-test results and sensitivity analyses amid COVID-19 uncertainty, impacting bank capital expectations.         | 2020-06-25      | JPM, BAC, WFC, C, XLF      |
| 18 | The FAA and Boeing began certification flights to evaluate proposed changes to the 737 MAX flight control system.                                      | 2020-06-29      | BA, GE, RTX, SPR           |
| 19 | China’s Hong Kong National Security Law was passed and took effect the same day, escalating geopolitical and market-access concerns.                   | 2020-06-30      | FXI, BABA, JD, TCEHY, AAPL |

Table 3: Index mapping for the 20 manually curated “idea” statements used in experiments (IDs 0–19), with orientative dates and example related tickers.

Table 4: Retrieving with topk\_mean ( $k = 1$ ) method vs the baseline TFIDF, for different companies (using company description queries).

| Company   | Mean #keyword<br>Embedding topk_mean ( $k = 1$ ) | #episodes keyword=0 | Mean #keyword<br>TFIDF (baseline) | #episodes keyword=0 |
|-----------|--------------------------------------------------|---------------------|-----------------------------------|---------------------|
| Apple     | 8.9                                              | 10                  | 9.7                               | 12                  |
| Google    | 13.52                                            | 3                   | 11.82                             | 16                  |
| Microsoft | 6.44                                             | 15                  | 7.5                               | 16                  |
| Amazon    | 15.46                                            | 3                   | 6.54                              | 21                  |

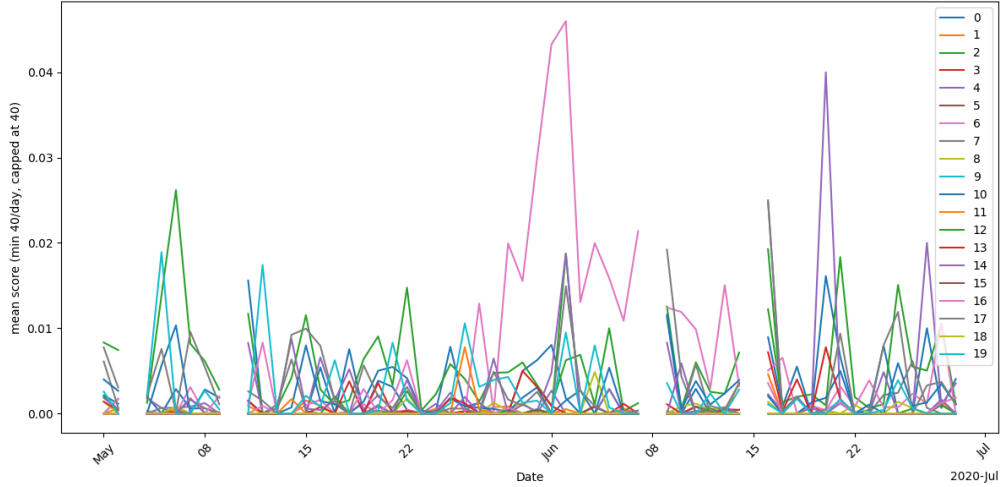


Figure 10: **Daily evolution of idea intensity for fine-grained ideas (May–June 2020).** Idea intensity is shown for the 20 fine-grained ideas (IDs 0–19; Appendix Table 3). For each episode, the idea score is computed using `pct_above(0.35)`, i.e., the fraction of transcript chunks whose cosine similarity to the corresponding idea-description embedding exceeds 0.35. Daily intensity is then computed as the mean of episode scores over a fixed 40-episode/day sample (days with fewer than 40 available episodes are excluded; when more than 40 exist, the daily sample is capped at 40).

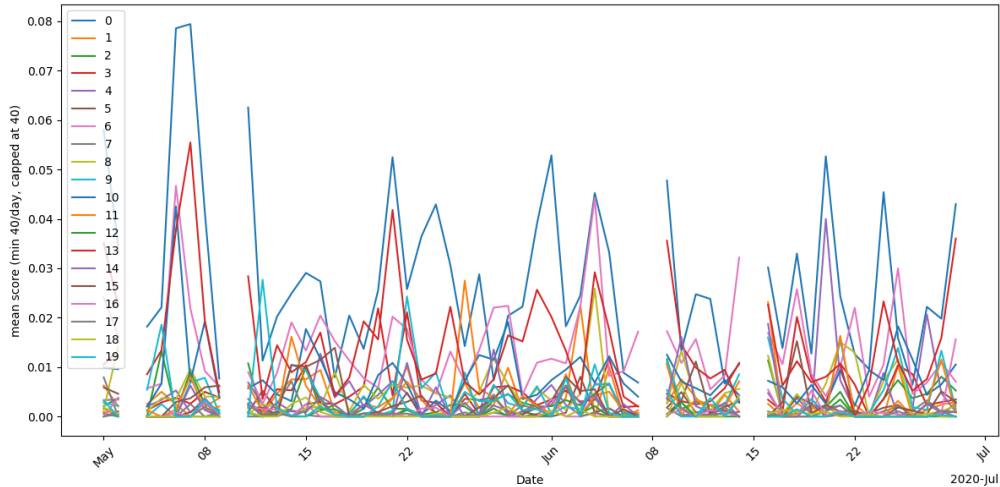


Figure 11: **Daily evolution of idea intensity for broader market-relevant ideas (May–June 2020).** Same computation pipeline as Figure, but applied to a set of broader ideas intended to capture themes plausibly relevant to stock movements. Episode-level scores are computed with `pct_above(0.35)` over chunks, and daily intensity is the mean episode score over a fixed 40-episode/day sample (minimum 40 required; capped at 40). Even if the trajectories appear similar, the absolute intensity values are typically nearly doubled because broader ideas are easier to match in transcript text.